सूक्ष्म , लघु और मध्यम उद्यम मंत्रालय
MINISTRY OF
MICRO, SMALL & MEDIUM ENTERPRISES

MSME SUSTAINABLE (ZED) CERTIFICATION

GUIDANCE DOCUMENT

NIC DIVISION 28 – Manufacture of Machinery and Equipment n.e.c

“This Guidance Document is meant to provide guidance to the MSMEs on the requirements of ZED Certification. Aim of the guidance document is to make the MSMEs aware on systems, processes and practices that may be adopted to effectively fulfil the requirements of the ZED Certification standard for the relevant NIC Division. The document will also assist the assessors to gain deeper insight into the sector specific requirements leading to uniformity in assessments.”

GUIDANCE FOR BRONZE CERTIFICATION

S. No.	Parameter	Guidance
1.	**Leadership**	Leadership should drive the organization towards growth by inculcating Zero Defect Zero Effect culture. Activities that may be included (but not limited to): ▪ Communicating & directing all stakeholders on the ZED culture/practices ▪ Periodically reviewing organizational performance on the ZED parameters including regulatory requirements
2.	**Swachh Workplace**	Identify and implement systems based on the requirements for cleanliness & hygiene that may include statutory & regulatory requirements, customer requirements, process and product requirements. Activities that may be included (but not limited to): ▪ Proper nomenclature and labelling of buildings/units/ yards/areas ▪ Task Identification based on the nature of task (housekeeping / process cleaning / maintenance etc.) ▪ Employee Role Assignment with well-defined responsibilities – who will do what ▪ Training of all employees on Cleanliness & Hygiene requirements ▪ Documentation / SOPs/ Log-sheets depicting activity schedules ▪ Ensuring proper Work Area Layout for process isolation / containment ▪ Periodic audits/checks and action plan in case of deviations, if any, identify Root Cause and take Corrective & Preventive Actions (CAPA) ▪ Tools like 5S may be implemented
3.	**Occupational (Workplace) Safety**	Identify and implement safety systems based on Hazard Analysis & Risk Assessment to identify safety requirements including statutory & regulatory requirements, customer requirements, process and product requirements. Activities that may be included (but not limited to): ▪ Establishing Safety Policy & communicating to all stakeholders including supply chain ▪ Training of all employees on safety and safety requirements ▪ Providing relevant Personal Protective Equipment (PPEs) to the employees/workers ▪ Signage & Pictograms/Pictures to alert about expected behavioral response, in adequate numbers ▪ Maintaining Accident/Incident register to record all incidents or accidents ▪ Inventory knowledge through categorize / classification of all materials (raw materials, products, process intermediates) according to their individual risk weightage ▪ Insulation & Warning Signage for cold & hot (glass wool / ceramic wool etc.) with descriptive signage ▪ Ventilation & Off-gas Management with a provision to monitor & maintain indoor air quality, as the case may be ▪ Periodic Hydraulic Testing for pressurized systems (boiler / vessels / jackets / coils / piping etc.)

S. No.	Parameter	Guidance
		▪ Provision of workers Amenities including personal protection kit, in-plant shower / eye wash; first-aid kit etc. ▪ Development of procedures to control hazards that may arise during non-routine operations (e.g., removing machine guarding during maintenance and repair) ▪ Systems to ensure timely availability of medical/emergency services ▪ Maintenance of emergency response equipment ▪ Mock drills for emergency evacuations in case of blasts, accidents or fire ▪ Periodic Safety audits and action plan in case of deviations and incidents, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
4.	**Measurement of Timely Delivery**	Identify and implement systems to measure the performance of timely delivery Activities that may be included (but not limited to): ▪ Recording contract delivery schedules ▪ Monitoring & measuring delivery timelines (On-Time in Full) ▪ Use of technology/systems to track and measure the deliveries
5.	**Quality Management**	Identify quality aspects and implement systems for products and processes to ensure consistent, high quality products and customer satisfaction. Activities that may be included (but not limited to): ▪ Establishing Quality Policy & communicating to all stakeholders ▪ Identifying quality requirements of the customers, regulators, products and processes and implement systems to fulfil the requirements ▪ Training all employees on quality & processes ▪ Ensuring that employees at all levels are involved in quality improvement initiatives ▪ Periodic Quality audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)

GUIDANCE FOR SILVER CERTIFICATION

<table>
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<tr><td>1.</td><td>Leadership</td><td>Leadership should drive the organization towards growth by inculcating Zero Defect Zero Effect culture. Activities that may be included (but not limited to):
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▪ Periodically reviewing organizational performance on the ZED parameters including regulatory requirements</td></tr>
<tr><td>2.</td><td>Swachh Workplace</td><td>Identify and implement systems based on the requirements for cleanliness & hygiene that may include statutory & regulatory requirements, customer requirements, process and product requirements. Activities that may be included (but not limited to):
▪ Proper nomenclature and labelling of buildings/units/ yards/areas
▪ Task Identification based on the nature of task (housekeeping / process cleaning / maintenance etc.)
▪ Employee Role Assignment with well-defined responsibilities – who will do what
▪ Training of all employees on Cleanliness & Hygiene requirements
▪ Documentation / SOPs/ Log-sheets depicting activity schedules
▪ Ensuring proper Work Area Layout for process isolation / containment
▪ Periodic audits/checks and action plan in case of deviations, if any, identify Root Cause and take Corrective & Preventive Actions (CAPA)
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▪ Establishing Safety Policy & communicating to all stakeholders including supply chain
▪ Training of all employees on safety and safety requirements
▪ Providing relevant Personal Protective Equipment (PPEs) to the employees/workers
▪ Signages & Pictograms/Pictures to alert about expected behavioral response, in adequate numbers
▪ Maintaining Accident/Incident register to record all incidents or accidents
▪ Inventory knowledge through categorize / classification of all materials (raw materials, products, process intermediates) according to their individual risk weightage
▪ Insulation & Warning Signage for cold & hot (glass wool / ceramic wool etc.) with descriptive signage
▪ Ventilation & Off-gas Management with a provision to monitor & maintain indoor air quality, as the case may be
▪ Periodic Hydraulic Testing for pressurized systems (boiler / vessels / jackets / coils / piping etc.)</td></tr>
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S. No.	Parameter	Guidance
		▪ Provision of workers Amenities including personal protection kit, in-plant shower / eye wash; first-aid kit etc. ▪ Development of procedures to control hazards that may arise during non-routine operations (e.g., removing machine guarding during maintenance and repair) ▪ Systems to ensure timely availability of medical/emergency services ▪ Maintenance of emergency response equipment ▪ Mock drills for emergency evacuations in case of blasts, accidents or fire ▪ Periodic Safety audits and action plan in case of deviations and incidents, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
4.	**Measurement of Timely Delivery**	Identify and implement systems to measure the performance of timely delivery Activities that may be included (but not limited to): ▪ Recording contract delivery schedules ▪ Monitoring & measuring delivery timelines (On-Time in Full) ▪ Use of technology/systems to track and measure the deliveries
5.	**Quality Management**	Identify quality aspects and implement systems for products and processes to ensure consistent, high quality products and customer satisfaction. Activities that may be included (but not limited to): ▪ Establishing Quality Policy & communicating to all stakeholders ▪ Identifying quality requirements of the customers, regulators, products and processes and implement systems to fulfil the requirements ▪ Training all employees on quality & processes ▪ Ensuring that employees at all levels are involved in quality improvement initiatives ▪ Periodic Quality audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
6.	**Human Resource Management**	Identify human resource requirements to ensure that the organizational goals are fulfilled as planned. Activities that may be included (but not limited to): ▪ Establishing skill set matrix for effective deployment of people ▪ Regularly training employees (both regular & contract) on relevant processes including quality, safety, energy, environment & natural resource conservation ▪ Reviewing the effectiveness of training & employee development initiatives ▪ Encouraging engagement of the employees through various initiatives that may include idea generation, suggestion box, quality circles, Kaizen, Six Sigma projects, Reward & recognition systems etc.
7.	**Daily Works Management**	Establish Daily Work Management System to ensure that Quality, Cost & Delivery, are measured, monitored & communicated. Activities that may be included (but not limited to): ▪ Setting up targets for Quality Cost & Delivery (QCD) for every day work ▪ Communicate the above to relevant employees ▪ Display QCD targets & trends at prominent locations for awareness and motivation

S. No.	Parameter	Guidance
		▪ Monitor the progress of QCD on daily basis to address deviations, if any, and identify Root Cause and take Corrective & Preventive Actions (CAPA), as required
8.	**Planned Maintenance & Calibration**	Identify the requirements of maintenance of all equipment & devices and develop systems for planned maintenance & calibration, as applicable. Activities that may be included (but not limited to): ▪ Scheduling maintenance & calibration of all equipment & devices, as required ▪ Adherence to the schedule ▪ Measuring Mean Time To Repair (MTTR) & Mean Time Before Failure (MTBF)
9.	**Process Control**	Identify the process control requirements to ensure that the requirements, as necessary, are fulfilled. Activities that may be included (but not limited to): ▪ Defining Process Flow diagram/ description of unit processes for the whole unit, section-wise, for each product ▪ Tagging to ensure clearly visible identification marking of each equipment, piping & instrumentation components in sync with PFD ▪ Identifying Critical Processes including welding, threading, fabrication, etc. based on process analysis, identification & setting up of boundary conditions of critical activities as may be appropriate ▪ Sensors and monitoring systems may include sensors, IoTs etc. to gather data/real-time data about for quality checks ▪ Periodic Quality audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
10.	**Product Quality & Safety (Testing / Certification)**	Identify the testing/certification requirements to ensure product quality & safety. Activities that may be included (but not limited to): ▪ Identifying requirements of testing including shock test, vibration test, etc. and/or certification for all products being manufactured e.g., testing for product specification, certification to relevant BIS standard/CE marking, etc. ▪ Checking the product(s) against specifications like item weight & dimensions, material & construction, item color, item marking & labelling etc. ▪ Provision of laboratory facility, wherever applicable ▪ Documentation / Log-sheets depicting analytical tests & related reports (raw materials, intermediates & products), following protocols laid down in concerned specifications, with traceability ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
11.	**Material Management**	Identify the requirements for material management to ensure optimal inventory, reduce material wastage and associated risks. Activities that may be included (but not limited to): ▪ Listing & Classification of materials being handled with indicative quantity, shelf life of material with expiry date, classification of inventory into slow moving, non- moving & obsolete inventory, etc. Also, the minimum inventory level should be defined so that material is procured at the right time and avoid shortage and stoppage in production

S. No.	Parameter	Guidance
		▪ Training of relevant employees and people on material management & handling ▪ Storage Facility in accordance with nature of the material, MSDS classification. Separate storage area should be assigned for raw materials, finished goods, quality rejected stock and expired stock. Flammable or explosive materials must be separated from other materials by a fire wall in the warehouse and kept in open space outside the warehouse. ▪ Materials identified as scrap should be stored separately and disposed-off periodically ▪ Inventory Control by recording & monitoring material(s) issued and received ▪ An efficient layout for warehouse should be maintained ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
12.	**Energy Management**	Ensure energy efficiency and conservation. Activities that may be included (but not limited to): ▪ Identification of energy sources that are utilized. Energy sources may be electricity, Diesel, Coal, Solar Power etc. ▪ Identification and monitoring of high energy consumption processes & equipment (boiler / chiller / TF heater etc.) to ensure optimum capacity utilization ▪ Establishing targets for energy savings ▪ Insulation in adequate thickness to minimize energy loss for cold & hot (glass wool / ceramic wool etc.) ▪ Selection & design of appropriate components/devices based on process requirement (viz., type of stirrer / pumps etc.) & BEE star rating equipment, wherever applicable, ▪ Installation of VFDs, LED lights etc. ▪ Layout of equipment & piping network to follow process flow for simplified material handling & optimum energy usage ▪ Electrical Systems to be designed for high Plant Load Factor (PLF), preference for 3 phase motors to avoid imbalance ▪ Implementation of automated control systems and building management system (IoT and sensor-based systems) to reduce uncertain breakdowns and shutdowns saving energy and bringing efficiency ▪ Training relevant employees & people ▪ Planning & adoption of renewable energy resources like Solar power ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
13.	**Environment Management**	Ensure compliance to environment management and pollution control requirements. Activities that may be included (but not limited to): ▪ Identifying environmental aspects & impacts of all product & process stages, for e.g., chemical fumes, wash residues, etc. ▪ Identifying and fulfilling all statutory & regulatory requirements including requisite environmental testing from NABL accredited laboratories

S. No.	Parameter	Guidance
		▪ Establishment of infrastructure (Effluent Treatment Plant (ETP), Sewage Treatment Plant (STP), Filters, Scrubbers, Cyclones etc.), equipment and systems to manage and monitor effluents (liquid discharge), emissions (gaseous discharge), hazardous waste, Noise pollution, e-waste, etc., as applicable ▪ Drainage Management for proper treatment / disposal mechanism of washing / effluents – separate channels for domestic & process/ wash discharge; plan to monitor & control ▪ Spill Management System to address spills, if any ▪ Paints and thinners, where used, should be properly disposed-off ▪ Training relevant employees & people ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
14.	**Measurement & Analysis**	Establish targets and process for measurement and analysis of the following (but not limited to) for improvement: ▪ Defects- arising during the process, inspection or at customer end ▪ Rework- arising during the process, inspection or at customer end ▪ Rejection- arising during the process, inspection or at customer end ▪ Cost of Poor Quality (COPQ)- overall cost of poor quality that includes, appraisal cost, prevention cost, internal cost & external cost ▪ Customer Satisfaction

GUIDANCE FOR GOLD CERTIFICATION

S. No.	Parameter	Guidance
1.	**Leadership**	Leadership should drive the organization towards growth by inculcating Zero Defect Zero Effect culture. Activities that may be included (but not limited to): ▪ Communicating & directing all stakeholders on the ZED culture/practices ▪ Periodically reviewing organizational performance on the ZED parameters including regulatory requirements
2.	**Swachh Workplace**	Identify and implement systems based on the requirements for cleanliness & hygiene that may include statutory & regulatory requirements, customer requirements, process and product requirements. Activities that may be included (but not limited to): ▪ Proper nomenclature and labelling of buildings/units/ yards/areas ▪ Task Identification based on the nature of task (housekeeping / process cleaning / maintenance etc.) ▪ Employee Role Assignment with well-defined responsibilities – who will do what ▪ Training of all employees on Cleanliness & Hygiene requirements ▪ Documentation / SOPs/ Log-sheets depicting activity schedules ▪ Ensuring proper Work Area Layout for process isolation / containment ▪ Periodic audits/checks and action plan in case of deviations, if any, identify Root Cause and take Corrective & Preventive Actions (CAPA) ▪ Tools like 5S may be implemented
3.	**Occupational (Workplace) Safety**	Identify and implement safety systems based on Hazard Analysis & Risk Assessment to identify safety requirements including statutory & regulatory requirements, customer requirements, process and product requirements. Activities that may be included (but not limited to): ▪ Establishing Safety Policy & communicating to all stakeholders including supply chain ▪ Training of all employees on safety and safety requirements ▪ Providing relevant Personal Protective Equipment (PPEs) to the employees/workers ▪ Signages & Pictograms/Pictures to alert about expected behavioral response, in adequate numbers ▪ Maintaining Accident/Incident register to record all incidents or accidents ▪ Inventory knowledge through categorize / classification of all materials (raw materials, products, process intermediates) according to their individual risk weightage ▪ Insulation & Warning Signage for cold & hot (glass wool / ceramic wool etc.) with descriptive signage ▪ Ventilation & Off-gas Management with a provision to monitor & maintain indoor air quality, as the case may be ▪ Periodic Hydraulic Testing for pressurized systems (boiler / vessels / jackets / coils / piping etc.)

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		▪ Provision of workers Amenities including personal protection kit, in-plant shower / eye wash; first-aid kit etc. ▪ Development of procedures to control hazards that may arise during non-routine operations (e.g., removing machine guarding during maintenance and repair) ▪ Systems to ensure timely availability of medical/emergency services ▪ Maintenance of emergency response equipment ▪ Mock drills for emergency evacuations in case of blasts, accidents or fire ▪ Periodic Safety audits and action plan in case of deviations and incidents, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
4.	**Measurement of Timely Delivery**	Identify and implement systems to measure the performance of timely delivery Activities that may be included (but not limited to): ▪ Recording contract delivery schedules ▪ Monitoring & measuring delivery timelines (On-Time in Full) ▪ Use of technology/systems to track and measure the deliveries
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		▪ Establishment of infrastructure (Effluent Treatment Plant (ETP), Sewage Treatment Plant (STP), Filters, Scrubbers, Cyclones etc.), equipment and systems to manage and monitor effluents (liquid discharge), emissions (gaseous discharge), hazardous waste, Noise pollution, e-waste, etc., as applicable ▪ Drainage Management for proper treatment / disposal mechanism of washing / effluents – separate channels for domestic & process/ wash discharge; plan to monitor & control ▪ Spill Management System to address spills, if any ▪ Paints and thinners, where used, should be properly disposed-off ▪ Training relevant employees & people ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
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15.	**Supply Chain Management**	Identify the requirements related to vendor/supply chain management. Activities that may be included (but not limited to): ▪ Developing processes to identify, select, evaluate & develop suppliers ▪ Monitoring performance of the vendors/ suppliers and out- sourced partners, periodically ▪ Timely communication to the vendors/suppliers for improvement and sensitizing them on energy, environment, natural resource management, best practices, etc.
16.	**Risk Management**	Identify the risks that may pose challenge/threat to the organization that may disrupt business. Activities that may be included (but not limited to): ▪ Identifying all the risks including safety and supply chain disruptions ▪ Developing plans to address the risks ▪ Implementing measures to mitigate/reduce risks ▪ Reviewing risks and mitigation plans periodically
17.	**Waste Management (Muda, Mura, Muri)**	Identify different types of wastes for eliminating/reducing them and enhance productivity. Activities that may be included (but not limited to): ▪ Training of employees on Muda, Mura, Muri ▪ Identifying Muda, Mura & Muri wastes ▪ Establishing targets & action plans to eliminate/reduce wastes ▪ Monitoring of Muda, Mura, Muri wastes
18.	**Technology Selection & Upgradation**	Understand the importance of technology in business operations. Activities that may be included (but not limited to):

S. No.	Parameter	Guidance
		▪ Periodic assessment of existing technology in the unit for need of upgradation, if any ▪ Training/exposure to the Best Available/latest Technology in the market ▪ Planning and adoption of latest technology including Internet of Things (IoT), sensors, robotics, cloud computing, Artificial Intelligence (AI), etc.
19.	**Natural Resource Conservation**	Ensure natural resource conservation. Activities that may be included (but not limited to): ▪ Identification of natural resources consumed (viz., water, biomass, fossil fuel etc.) ▪ Assessment of possibility to reduce consumption & / or, recovery of such resources from process effluents (viz., water through ZLD approach) ▪ Establishing targets for natural resource conservation ▪ Workplace Design to maximize utilization of natural light, as may be appropriate ▪ Ventilation / HVAC Design to promote natural air draft, as may be appropriate ▪ Adopting measures and promoting water conservation by reducing water usage and recycling of the water in the processes ▪ Training relevant employees & people ▪ Periodic audits and action plan in case of deviations, if any, to identify Root Cause and take Corrective & Preventive Actions (CAPA)
20.	**Corporate Social Responsibility**	Ensure that the organization operates in a socially responsible manner and contributes to Societal well-being. Activities that may be included (but not limited to): ▪ Identifying statutory & regulatory requirements, as applicable ▪ Identifying areas that the organization wants to contribute towards the societal well-being. These may include, but are not limited to Organizational Governance, Labor Practices, Environment, Fair Operating Practices, Community Involvement & Development ▪ Establishing targets & resources ▪ Monitoring & reviewing the progress